

Homework 2

Due: Mar. 30th

Turn in your homework in class

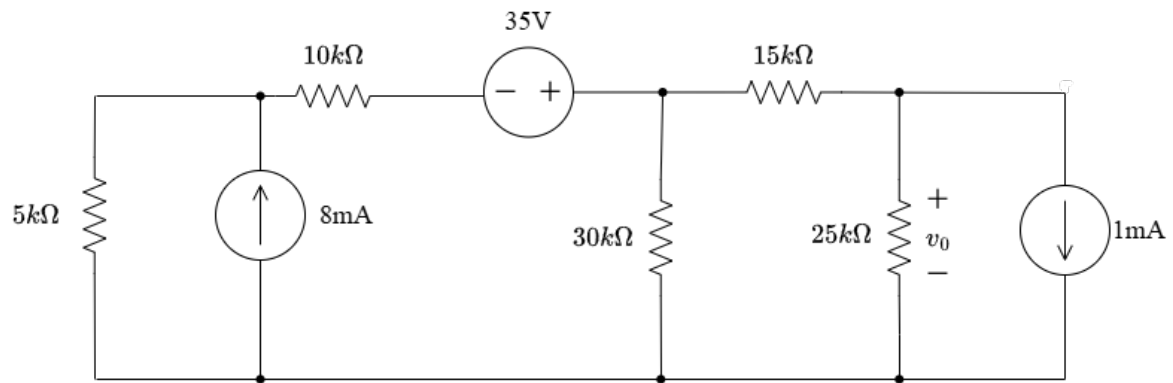
Rules

- Please work on your own. Discussion is permissible, but extremely similar submissions will be judged as plagiarism.
- Please show all intermediate steps: a correct solution without an explanation will get zero credit.
- Please submit on time. No late submission will be accepted.
- Please prepare your submission in English only. No Chinese submission will be accepted.
- Please retain 3 decimal places if rounding is required (unless otherwise specified).

1 Linearity Principle & Superposition Principle

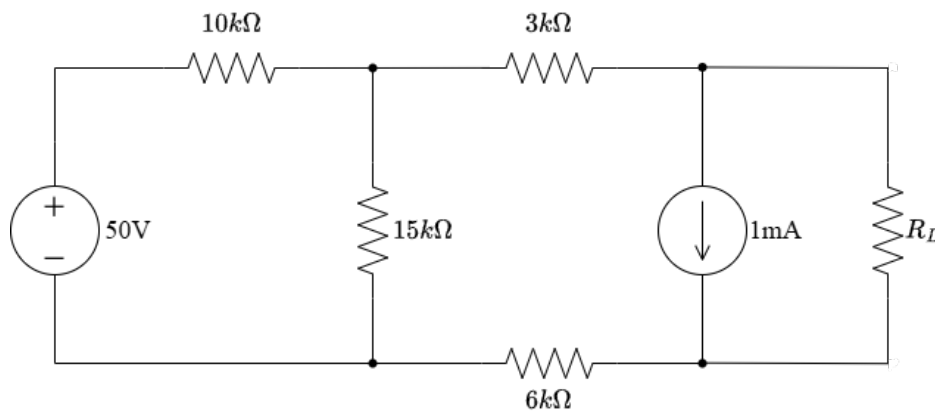
[15 points]

- (a) Use the superposition principle to find the voltage v_0 in the circuit.
- (b) Verify your solution using the mesh analysis.



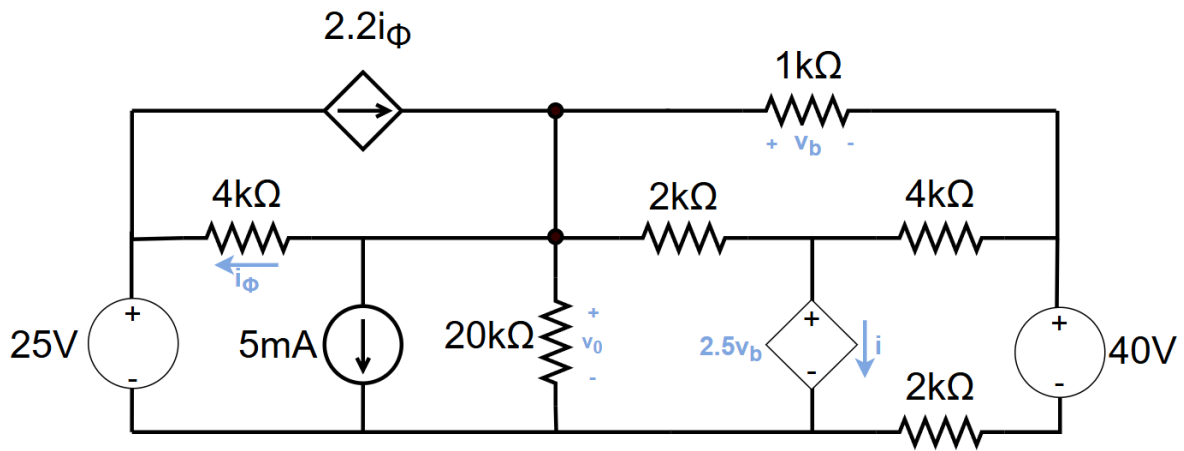
2 Maximum Power Transfer

[10 points] Determine the value of R_L that results in maximum power transfer to R_L , and calculate that maximum power.



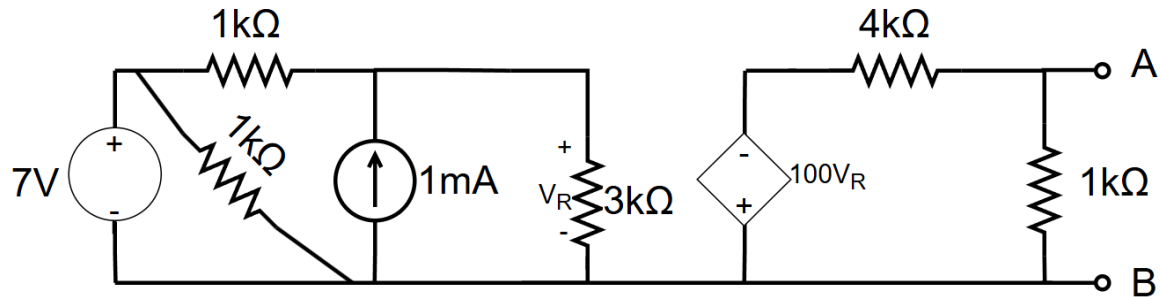
3 Superposition Principle

[13 points] Use the superposition theorem to find v_o and i in the circuit.



4 Thevenin's Theorems & Norton's Theorems

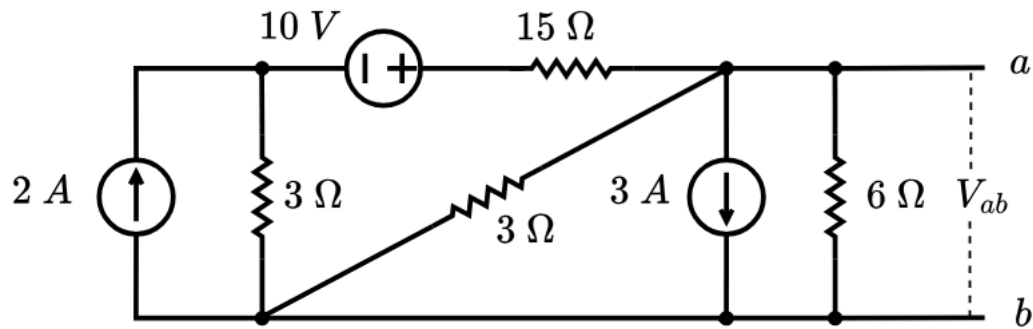
[12 points] Given the circuit below, find the Thevenin equivalent circuit and the Norton equivalent circuit at terminals A and B respectively.



5 Source Transformation & Norton's Theorems

[12 points]

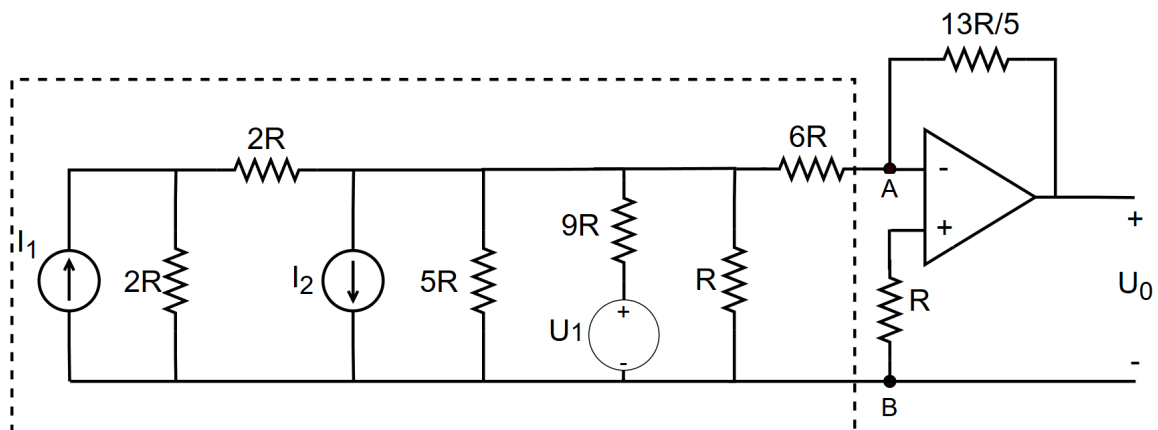
- (a) Use source transformation to find V_{ab} . [6 points]
- (b) Find the Norton equivalent circuit at terminals a and b. [6 points]



6 Thevenin's Theorems & Op Amp

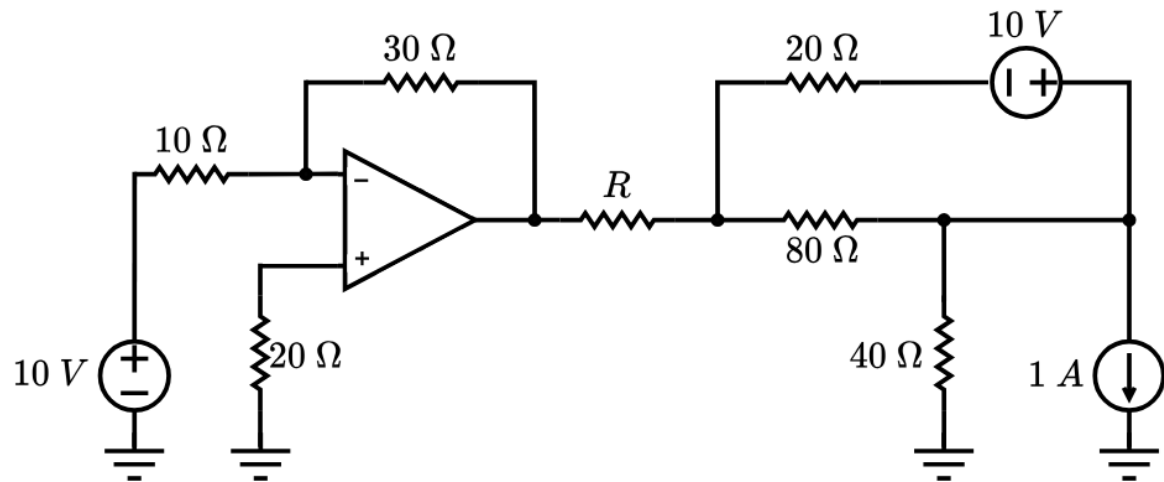
[14 points] We suggest that I_1, I_2, U_1 and R are already known. Assume that the operational amplifier is ideal and operates in the linear region.

- Find the Thevenin equivalent circuit of the part framed by the dotted line (i.e. find the Thevenin equivalent circuit at terminal A and B). [9 points]
- Express the output voltage U_0 in terms of I_1, I_2 and U_1 . [7 points]



7 Maximum Power Transfer

[14 points] Assume the operational amplifier is ideal and operates in linear region. Then find the load R for maximum power consumption and the maximum power consumption of it.



8 Op Amp

[10 points] Assume the operational amplifier is ideal and operates in linear region. Then find V_{o1} and V_{o2} in terms of V_i .

